PackLab: reproducible Percus–Yevick correlations for hard-sphere mixtures

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Abstract

PackLab is an open-source Python package centred on the analytical Percus–Yevick (PY) solution for multicomponent hard-sphere mixtures. It evaluates equilibrium-reference partial pair correlations and structure factors with unit-aware inputs and explicit Fourier-resolution diagnostics. A comparison with digitised PY curves from Tsang et al. demonstrates the matched binary-mixture workflow. Compiled random sequential addition (RSA) and Metropolis Monte Carlo (MC) tools are included as complementary add-ons for constructing non-overlapping configurations and sampling fixed-volume hard-sphere states; neither is treated as interchangeable with the analytical model.

1 Statement of need

Hard-sphere-mixture correlation functions and structure factors are needed as reproducible structural inputs for colloids, particulate materials, porous media, disordered optical systems, and related scattering models [5, 8]. Researchers need to vary radius distributions, compositions, and volume fractions without silently mixing incompatible physical inputs, units, or Fourier-grid settings across disconnected scripts.

Dependent-scattering calculations for dense, polydisperse suspensions provide a concrete example of this need. Attendu et al. combine single-particle Mie theory with a multicomponent PY hard-sphere model to predict the scattering properties of Intralipid across wavelength and concentration [1]. Such applications require the particle-size distribution, mixture correlation model, and numerical settings to remain internally consistent.

PackLab provides this analytical reference in one documented Python interface. Its core `analytical` namespace evaluates the PY hard-sphere-mixture model [7, 6], retaining the mixture definition, dimensional inputs, and Fourier-resolution choices with the result. The package is therefore suited to analytical parameter sweeps and structure-aware scattering models. Complementary RSA and MC workflows provide explicit configurations for downstream coordinate-based calculations and carefully interpreted comparisons; they are not substitutes for the equilibrium analytical reference.

2 Scientific scope

2.1 Percus–Yevick hard-sphere mixtures

The analytical workflow evaluates the multicomponent PY approximation for hard-sphere mixtures. `PercusYevickDomain` converts radii, number fractions, and total volume fraction into species densities. `PercusYevickSolver` then evaluates partial pair correlations $g_{ij}(r)$ and structure factors on a user-supplied or automatically constructed wavenumber grid.

PY is an approximate equilibrium reference. Its approximation error depends on mixture density, composition, and size ratio, while its numerical error depends separately on the inverse-transform grid. PackLab exposes both choices: automatic grids provide a documented default and the grid-construction interface permits explicit resolution studies. The digitised binary-mixture PY curves of Tsang et al. [9], used in Figure 3, provide an external visual comparison at matched radii and partial volume fractions.

2.2 Configuration-generation add-ons

2.2.1 Random sequential addition

RSA proposes one sphere at a time. For a candidate centre $\mathbf{x}_i$ and radius a_i, PackLab accepts the candidate only if

$$\|\mathbf{x}_i - \mathbf{x}_j\| \geq a_i + a_j \tag{1}$$

for every previously accepted sphere j. Periodic simulations use the minimum-image convention. For a domain with volume V, the realised packing fraction is

$$\phi = \frac{1}{V} \sum_i \frac{4\pi a_i^3}{3}. \tag{2}$$

RSA is irreversible and history dependent: once a sphere is accepted, it remains fixed for the rest of the run, while rejected candidates are discarded and never reconsidered. Consequently, early accepted spheres can block later positions permanently, so the final configuration depends on the order of random proposals rather than being a sample from the equilibrium hard-sphere fluid [3]. PackLab makes this protocol explicit through independent stopping conditions, a stored random seed, periodic or finite domains, and constant, uniform, normal, log-normal, and discrete radius samplers. A spatial-neighbour grid limits overlap checks to nearby accepted spheres, preserving the RSA acceptance rule while avoiding an all-pairs test for every proposal.

2.2.2 Metropolis Monte Carlo

The MC add-on takes a valid non-overlapping configuration, holds its particle count, radii, and class labels fixed, and proposes symmetric single-particle displacements. For hard spheres, a symmetric proposal is accepted exactly when it remains within the domain and introduces no overlap. This Markov chain targets the uniform equilibrium distribution over all valid configurations at the specified particle count and volume, but its early states still reflect the chosen initial configuration. Users therefore discard an initial *burn-in* interval: enough proposed moves for the chain to lose this initial-state memory. Successive saved states can also remain similar; their *autocorrelation* measures that similarity and determines how far apart samples should be taken before they are treated as approximately independent. Finally, a simulation contains a finite box and finite number of particles, so measured correlations can change with box size or particle number; checking these *finite-size effects* establishes whether the result represents the intended bulk system. RSA can provide a convenient initial state, but its irreversible deposition history is not part of the subsequent MC target distribution once the chain has equilibrated. The polydisperse hard-sphere Monte-Carlo structure factors of Frenkel et al. [4] remain a candidate higher-precision equilibrium benchmark, while the ensemble and physical inputs of Ding et al. [2] would need to be matched before comparison.

3 Software design and functionality

Performance-critical analytical and configuration kernels are implemented in C++20 and exposed as native Python extensions. The Python layer provides NumPy-compatible results, `TypedUnit`-based dimensional inputs, plotting helpers, type information, and executable examples. Native extensions are installed beside their public Python APIs, keeping imports such as `PackLab.monte_carlo` and `PackLab.analytical` stable and discoverable.

An analytical workflow consists of a `PercusYevickDomain` and `PercusYevickSolver`. Beginner-facing examples use `wavenumber="auto"`, which selects a zero-inclusive grid from the smallest particle radius and requested distance range. Users can instead supply a grid with `make_wavenumber_grid` when they need explicit control over radial resolution and Fourier sampling.

For explicit configurations, an RSA workflow combines a `PackingDomain`, a radius sampler, and `RSAOptions`; `RSASimulator.run()` returns a `PackingResult`. The result contains centres, radii, class labels, packing statistics, and total or partial pair-correlation estimates. `MetropolisSimulator` accepts such a configuration with `MetropolisOptions` and reports displacement-move acceptance statistics. For RSA ensemble calculations, `PackingEstimator` reports mean and standard deviation of partial correlations and insertion, rejection, runtime, and realised-packing diagnostics.

4 Complementary configuration workflows

Figure 2 shows a fixed-seed periodic RSA realisation of a binary mixture. The cross-section provides a direct geometric diagnostic of one generated configuration, while the correlation panel reports ensemble statistics for the same mixture. The example is generated entirely from the public API by `docs/manuscript/figures/create_figure2.py`. The other quantitative figures are generated by `create_figure3.py` and `create_figure4_independent_validation.py` in the same directory; their digitised Tsang reference data are versioned in `docs/manuscript/data/`. Table 1 collects the physical, stochastic, and numerical controls for all quantitative figures.

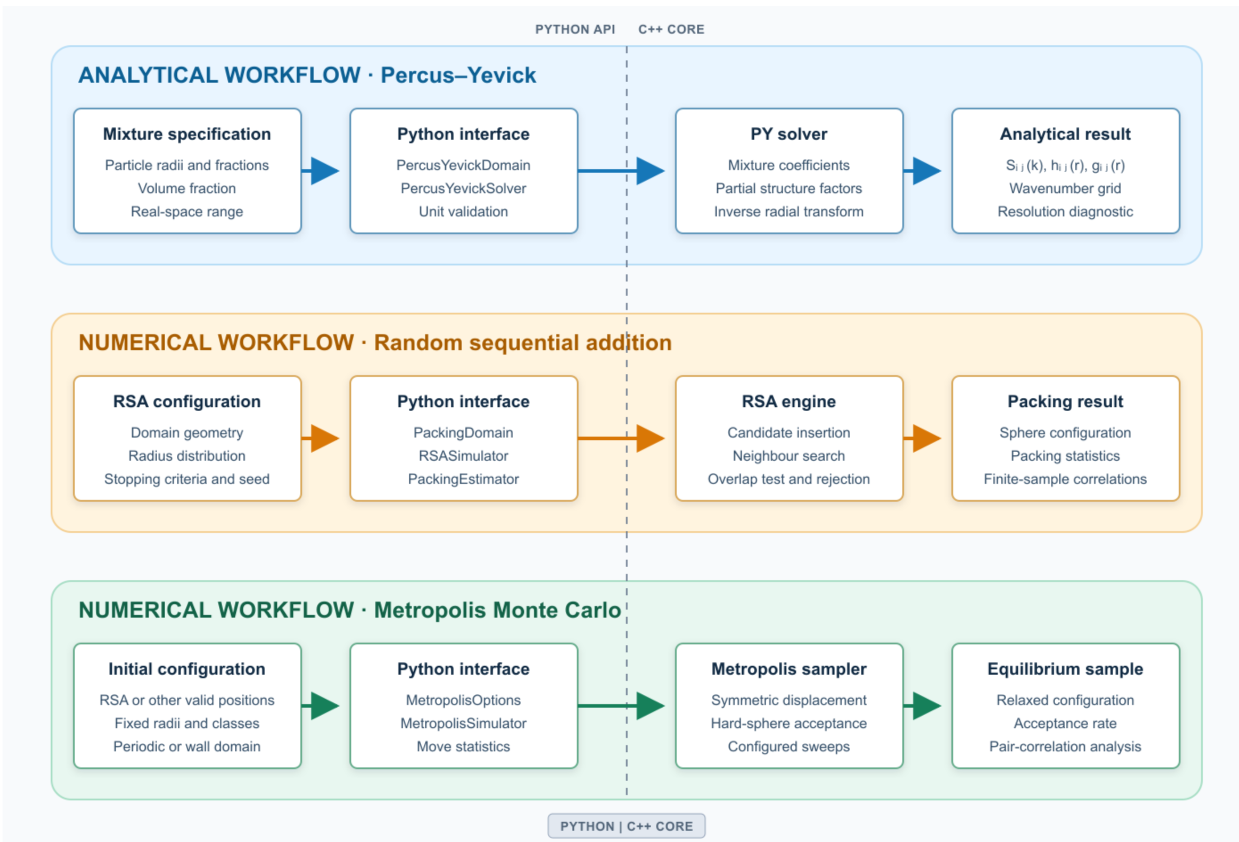

Figure 1: PackLab centres the PY equilibrium hard-sphere-mixture workflow and provides complementary numerical workflows. The analytical workflow evaluates pair correlations and structure factors for specified mixture inputs. The RSA and MC workflows create, respectively, irreversible deposition configurations and fixed-volume equilibrium samples. All workflows have unit-aware Python APIs backed by native C++ kernels, but their results are compared and interpreted rather than treated as interchangeable.

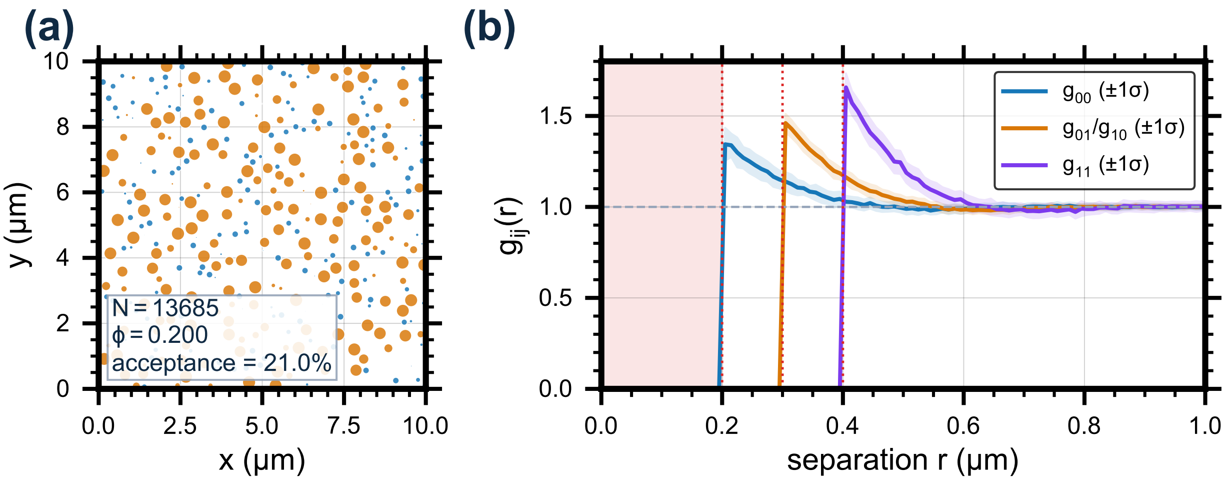

Figure 2: Reproducible periodic RSA demonstration for a two-radius mixture. The cubic domain has side length 10 μm; sphere radii are 100 and 200 nm with requested number fractions 0.65 and 0.35; the target packing fraction is 0.20; and the random seed is 20260828. (a) Central cross-section of the accepted three-dimensional configuration, with colour indicating radius class. (b) The unique partial pair correlations, $g_{00}(r)$, $g_{01}(r) = g_{10}(r)$, and $g_{11}(r)$, estimated from 64 independent RSA realisations. The legend identifies each ensemble-mean coloured line; its matching same-colour band shows $\pm 1\sigma$, where σ is the sample standard deviation across realisations. The red shaded interval is the exclusion range shared by all pairs, while the red dotted lines mark the three contact separations, 0.20, 0.30, and 0.40 μm. The dashed horizontal line marks the uncorrelated limit $g_{ij}(r) = 1$.

5 Percus–Yevick validation and numerical resolution

5.1 Analytical correlation and structure-factor conventions

For the RSA estimator, used only for the complementary finite-configuration comparison below, $g_{ij}(r_b)$ is the ratio between the observed ordered i–j pair count in radial shell b and the ideal-gas expectation for the same configuration. For species counts N_i and N_j, domain volume V, and shell volume ΔV_b, the latter is

$$E_{ij,b} = N_i\,(N_j - \delta_{ij})\,\frac{\Delta V_b}{V}, \qquad g_{ij}(r_b) = \frac{C_{ij,b}}{E_{ij,b}}, \tag{3}$$

where δ_{ij} is the Kronecker delta and $C_{ij,b}$ is the observed ordered count. The radial bins are spherical shells with centres $r_b = (b+1/2)\Delta r$ and extend to half the shortest box dimension; periodic estimates use minimum-image separations. When pair sampling is requested, the observed histogram is rescaled by the ratio of all unordered pairs to examined unordered pairs before this normalization. For the analytical workflow, $h_{ij}(r) = g_{ij}(r) - 1$ and $H_{ij}(k) = \sqrt{\rho_i \rho_j}\,\tilde{h}_{ij}(k)$ is the density-scaled reciprocal-space total correlation. We use the corresponding density-scaled partial structure-factor convention

$$S_{ij}(k) = \delta_{ij} + H_{ij}(k). \tag{4}$$

This convention makes both the pair-correlation and structure-factor comparisons dimensionless and identifies the density factors required when transforming between them.

The PY calculation uses an inverse radial Fourier transform, so its real-space result depends on wavenumber-grid extent and sampling density. For an isotropic mixture, the real-space total correlation is recovered from $H_{ij}(k)$ through

$$h_{ij}(r) = \frac{1}{2\pi^2 \sqrt{\rho_i \rho_j}} \int_0^\infty k^2 H_{ij}(k)\,\mathrm{sinc}(kr)\,\mathrm{d}k. \tag{5}$$

At the maximum requested distance $r_{\max}$, the $\mathrm{sinc}(kr)$ factor has wavenumber period $2\pi/r_{\max}$, which is the most rapidly varying kernel over the requested distance range. A truncated grid omits high-wavenumber content, whereas a coarse grid under-samples oscillations in the sinc kernel. Either case can produce a visually smooth but numerically misleading correlation function.

For `wavenumber="auto"`, PackLab uses a real-space resolution of one twentieth of the smallest radius and twelve samples per sinc-kernel oscillation at the maximum requested distance. When a user-provided grid has fewer than eight samples per oscillation, PackLab raises a runtime warning. These are numerical defaults, not physical constants, and the grid-construction API allows stricter choices.

5.2 External binary-mixture comparison

Figure 3 is the manuscript's primary external comparison of the PY implementation. It compares PackLab's mono- and binary-mixture solutions with manually digitised PY curves from Tsang et al. [9] at matched volume fractions and radius ratio. The expected hard-core contact locations, decay toward the uncorrelated limit, and damped post-contact structure agree visually across both monodisperse curves and all three binary partial correlations. At the digitised sample locations, the per-curve root-mean-square (RMS) differences are 0.174 and 0.250 for the monodisperse $f = 0.2$ and $f = 0.3$ curves, and 0.027, 0.034, and 0.088 for g_{11}, g_{12}, and g_{22} of the binary mixture, respectively. The figure and these statistics are therefore reproducible implementation-comparison evidence, rather than high-precision error measurements, because the reference was digitised from printed plots.

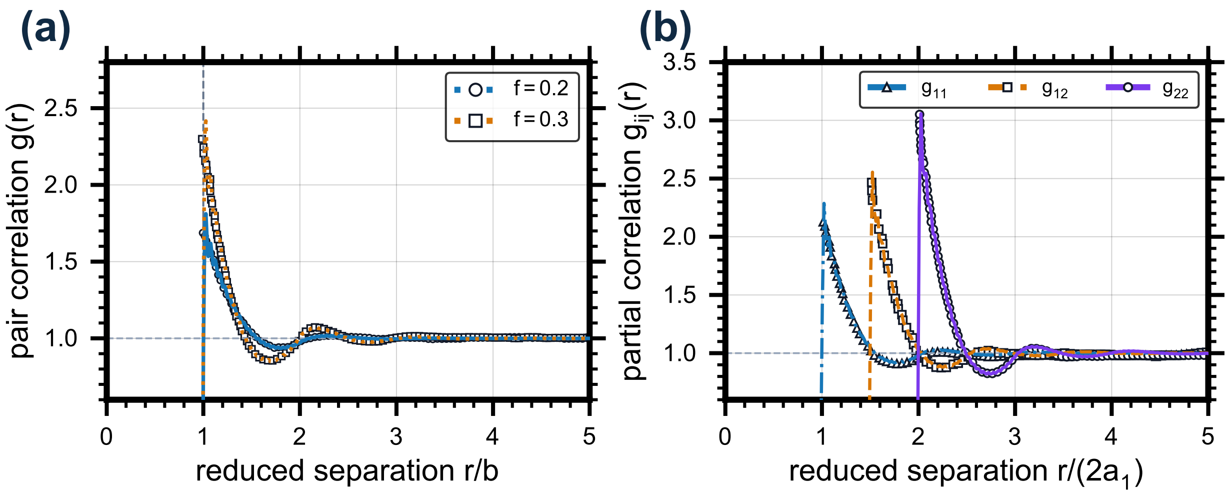

Figure 3: PackLab PY pair correlations (coloured lines) and digitised reference curves from Tsang et al. [9] (open markers). (a) Monodisperse correlations at volume fractions $f = 0.2$ and $f = 0.3$, plotted against r/b. (b) Binary-mixture partial correlations for $a_2 = 2a_1$ and partial volume fractions $f_1 = 0.2$, $f_2 = 0.1$, plotted against $r/(2a_1)$.

5.3 Complementary RSA comparison and numerical convergence

Figure 4 distinguishes model comparison from numerical verification. Panel (a) compares all unique partial correlations from a 100-realisation RSA ensemble with PY evaluated for the same binary-mixture inputs and the same radial coordinates. It demonstrates how PackLab makes the common inputs, statistical uncertainty, and model-dependent differences inspectable. The differences in panel (a) are expected physical consequences of comparing finite, irreversible RSA with an equilibrium PY reference, rather than a numerical-error measure. Panel (b) instead isolates numerical convergence by measuring the PY result against a separately specified, much finer wavenumber grid.

6 Quality control, availability, and limitations

PackLab includes public-API tests for analytical hard-core exclusion, automatic-grid selection, grid warnings, and convergence against a refined reference, as well as samplers, packing validity, periodic boundaries, and Metropolis move validation. Executable documentation examples are organised around analytical, RSA, MC, and comparison workflows. Continuous integration (CI) builds the native extensions and runs the test suite for CPython 3.11–3.13.

PackLab version 0.1.0 requires Python 3.10 or later, is released under the MIT License, and is available from Python Package Index (PyPI) and conda. Source code, documentation, issue tracking, and release history are available at `https://github.com/MartinPdeS/PackLab`. Release-specific citation metadata is maintained in `.zenodo.json`. The repository's v0.1.0 release is archived at 10.5281/zenodo.20207810.

The current scope is three-dimensional, spherical, hard-core particles in cuboidal domains. RSA should not be interpreted as an equilibrium-fluid simulator; MC requires explicit equilibration assessment; and PY should not be interpreted as a measured structure or a general interaction model.

The practical limits of both workflows depend on the requested mixture and numerical settings. For

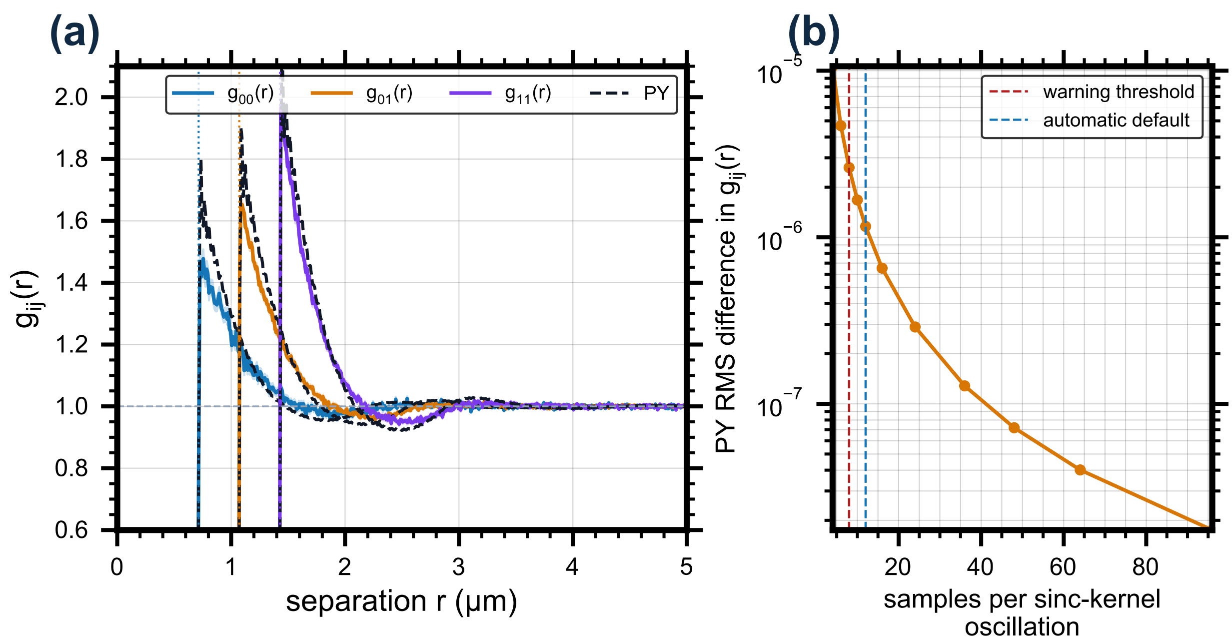

Figure 4: Matched RSA–PY comparison and PY grid convergence for a binary mixture with radii 0.357 and 0.714 µm, equal number fractions, and total volume fraction 0.25. (a) RSA ensemble means (coloured solid curves) with one-standard-error bands and PY references (black dashed curves) for the three unique partial correlations; dotted lines mark hard-core contact distances. (b) RMS difference from a 512-samples-per-oscillation PY reference as the wavenumber grid is refined; red and blue dashed lines mark the warning threshold (8) and automatic default (12), respectively.

Table 1: Reproducibility controls for the quantitative figures. Figure numbers refer to the rendered manuscript.

Figure	Mixture and geometry	Stochastic controls	Numerical and reference controls
2	Radii 100 and 200 nm; number fractions 0.65 and 0.35; periodic cube of side 10 μm; target $\phi = 0.20$.	Seed 20260828; 64 independent realisations; 500 radial bins.	Pair correlations to 2 μm; maximum attempts and consecutive rejections of 100000 and 20000.
3	Monodisperse $f = 0.2, 0.3$; binary radii $a_1 = 1$, $a_2 = 2$ μm with $f_1 = 0.2$, $f_2 = 0.1$.	Deterministic analytical calculations.	`wavenumber="auto"`; digitised Tsang et al. curves; see text for RMS differences.
4	Radii 0.357 and 0.714 μm; equal number fractions; $\phi = 0.25$; periodic cube of side 30 μm.	Seed 2026; 100 realisations; 1500 radial bins.	Grid resolutions 4–96 samples per oscillation; 512-samples-per-oscillation reference; $r_{\max} = 10$ μm and 1000 radial points.

RSA, insertion becomes progressively less efficient as the requested packing fraction increases; finite domains, broad radius distributions, and stringent stopping conditions can prevent the requested target from being reached. Consequently, users should report the realised packing fraction, stopping diagnostics, random seed, ensemble size, and domain size rather than treating a requested target as an achieved state. Pair-correlation estimates from finite configurations additionally depend on binning and are subject to finite-size uncertainty, particularly near the largest separations accessible in a domain. For MC, users should assess burn-in, autocorrelation, proposal acceptance, and finite-size effects before treating separated chain states as equilibrium samples.

For PY, the approximation error depends on density, composition, and particle-size ratio, and is separate from Fourier-grid discretisation error. The automatic grid and under-resolution warning are useful safeguards, but they do not establish convergence for every mixture, especially when near-contact structure is sharp, size ratios are extreme, or correlations are required over a long distance range. Users should refine the wavenumber grid and compare against an independently chosen equilibrium benchmark when the analytical result informs a quantitative conclusion. PackLab's separate namespaces, explicit input definitions, recorded stochastic settings, and grid diagnostics are designed to make these limitations visible at the point of use.

7 Acknowledgements

The author thanks the open-source scientific Python ecosystem and the users who provide feedback, issues, and reproducible examples.

References

[1] Xavier Attendu, Martin Poinsinet de Sivry-Houle, Mona Shahsavari, D. Faber, Edwin van der Pol, and Ton G. van Leeuwen. Scattering properties of Intralipid modeled using dependent scattering theory. *Journal of Biomedical Optics*, 31(8):085003, 2026.

[2] K. H. Ding, C. E. Mandt, L. Tsang, and J. A. Kong. Monte carlo simulations of pair distribution functions of dense discrete random media with multiple sizes of particles. *Journal of Electromagnetic Waves and Applications*, 6(7):1015–1030, 1992.

[3] Joseph Feder. Random sequential adsorption. *Journal of Theoretical Biology*, 87(2):237–254, 1980.

[4] D. Frenkel, R. Vos, C. G. de Kruif, and A. Vrij. Structure factors of polydisperse systems of hard spheres: A comparison of monte carlo simulations and percus–yevick theory. *The Journal of Chemical Physics*, 84:4625–4630, 1986.

[5] Jean-Pierre Hansen and Ian R. McDonald. *Theory of Simple Liquids: With Applications to Soft Matter*. Academic Press, 4 edition, 2013.

[6] Joel L. Lebowitz. Exact solution of generalized percus–yevick equation for a mixture of hard spheres. *Physical Review*, 133(4A):A895–A899, 1964.

[7] Jerome K. Percus and George J. Yevick. Analysis of classical statistical mechanics by means of collective coordinates. *Physical Review*, 110(1):1–13, 1958.

[8] Salvatore Torquato. *Random Heterogeneous Materials: Microstructure and Macroscopic Properties*. Springer, 2002.

[9] Leung Tsang, Jin Au Kong, Kung-Hau Ding, and Chi On Ao. *Scattering of Electromagnetic Waves: Numerical Simulations*. John Wiley & Sons, 2001.